



Genome-wide analysis of the mitogen-activated protein kinase gene family in *Solanum lycopersicum*

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ABSTRACT

Mitogen-activated protein kinases (MAPKs), a family of Ser/Thr protein kinases, play an essential role in mediating biotic and abiotic stress responses in plants. In this study, we investigated 16 putative *SIMAPK* genes from tomato genome and compared them with those from *Arabidopsis*. The full cDNA sequences of 13 novel *SIMAPKs* in tomato were obtained through PCR amplification. A comprehensive genome-wide analysis of *SIMAPKs* in tomato is presented, including their gene structure, phylogeny, genome localization, and expression profiles. Phylogenetic analysis of the 16 *SIMAPKs* and 20 *AtMAPKs* from *Arabidopsis* indicated that the *SIMAPK* genes were clustered into four major groups, and genes within the same groups had similar exon–intron structures. All *SIMAPK* proteins in groups A, B and C had a Thr–Glu–Tyr (TEY) activation domain, whereas those in group D contained a Thr–Asp–Tyr (TDY) activation domain. The analysis of 5′-upstream region of *SIMAPKs* revealed a group of putative *cis*-acting elements related to stress responses. Expression analysis of *SIMAPK* genes using RT-PCR and real-time quantitative PCR demonstrated that all *SIMAPK* transcripts were able to be detected in at least one investigated tissue, and some of them exhibited tissue-specific expression patterns. The transcript abundance of nearly all *SIMAPK* genes was increased in response to heat stress treatment. Our data provided an insight into the evolution of the gene family and a useful reference for further functional analysis of *MAPK* family genes in tomato.

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1. Introduction

Plants are often exposed to ever-changing external stimuli during their growth and development, including pathogen infection, drought, salinity and various other biotic and abiotic stresses. These extracellular stimuli are perceived by the plant through a highly elaborate signaling network, which leads to an adaptive stress response at cellular and organismal levels. A specific class of serine/threonine protein kinases, mitogen-activated protein kinases (MAPKs), play a vital role in the transduction of various extra- and intracellular signals and are conserved throughout eukaryotes (Rohila and Yang, 2007). The MAP kinases are components of a linear cascade of three consecutively acting protein kinases: MAPKK kinases (MAPKKKs), MAPK kinases (MAPKKs), and MAPKs (Jonak and Okresz, 2002). MAPKs are activated when both tyrosine and threonine residues in the TXY/TDY motif are phosphorylated by dual-specificity kinases, MAPKKs, which in turn are activated by

MAPKKKs via phosphorylation on two serine/threonine residues in the conserved Thr/Ser motif (Zhang et al., 2006).

MAPK contains eleven domains (I–XI) that are found in all serine/threonine protein kinases. Threonine and tyrosine residues existing between domains VII and VIII of MAPKs are well conserved. They have to be phosphorylated for the activation of the MAPKs (Hirt, 1997). The high degree of sequence similarity of MAPKs might reflect their functional conservation among different plant species. In all plant species, MAPKs carry either a Thr–Glu–Tyr (TEY) or Thr–Asp–Tyr (TDY) phosphorylation motif at the active site. In contrast to TEY MAPKs, all TDY MAPKs have long C-terminal extensions. To date, functional data for TDY MAPKs is only available for one rice MAPK, BWMK1 (blast- and wound-induced MAP kinase). By contrast, the TEY MAPKs have been studied in many plant species including *Arabidopsis*, Medicago, tobacco, tomato, parsley as well as rice (Cheong et al., 2003).

MAPKs regulate diverse cellular programs by relaying extracellular signals to intracellular responses (Cargnello and Roux, 2011). There is increasing evidence that plant MAPKs play an important role in the response to a broad variety of biotic and abiotic stresses including wounding, pathogen infection, temperature, drought, salinity and plant hormones (Lee et al., 2011; Romeis, 2001; Tena et al., 2001; Zhang and Klessig, 2001). The *Arabidopsis* genome contains 20 MAPK genes (Ichimura et al., 2002) and much experimental effort has been devoted

Abbreviations: At, *Arabidopsis thaliana*; MAPK, mitogen-activated protein kinase; ORF, open reading frame; RT-PCR, reverse transcription PCR; QRT-PCR, quantitative real-time PCR; Sl, *Solanum lycopersicum*; SOL, The International Solanaceae Project.

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to determining the biological processes that are regulated by each of these kinases. To date, the most extensively studied MAPKs are MPK3, MPK4 and MPK6 in *Arabidopsis* and crop plants, which are mainly involved in stress responses (Ichimura, et al., 2000; Jonak et al., 2002). In tobacco, ozone induces the activation of a p46 MAPK within minutes (Samuel et al., 2000). It has been proven that the activation of salicylate-induced protein kinase (SIPK), the major ROS-induced MAPK, was important for the effective control of ROS damage, and an unexpected interplay between the activities of SIPK and WIPK (wound-induced protein kinase) has been reported (Samuel and Ellis, 2002). Although the ozone-induced activation of MPK3 and MPK6 in *Arabidopsis* was independent of ethylene, it was dependent on salicylic acid and resulted in the nuclear translocation of MAPKs (Ahlfors et al., 2004). Several MAPKs have been shown to be activated by osmotic stresses or NaCl in *Medicago* and tobacco (Jonak et al., 2002). MPK4 and MPK6 in *Arabidopsis* were phosphorylated and activated by MKK2 which was specifically activated by cold and salt stress (Teige et al., 2004).

Several MAPK genes have been found to play an important role in the regulation of developmental processes, specifically in the flower development (Asai et al., 2002; Boudsocq and Lauriere, 2005; Nishihama et al., 2001; Rohila and Yang, 2007). The majority of stamens in *mpk3/+ mpk6/-* mutants do not form normal cell types. Hord et al. (2008) suggested that *MPK3/MPK6* genes were key regulators of cell proliferation and differentiation during anther development. A tobacco MAP kinase p45^{ntf4} was identified and proven to be involved in pollen maturation and germination and be activated by hydration during pollen germination (Wilson et al., 1997). A pea MAP kinase gene, *PsMAPK3*, was induced by gibberellic acid and 6-benzyladenine in unpollinated pea ovaries, which suggested that gibberellins and cytokinins might activate fruit set in pea through a pathway mediated by *PsMAPK3* (Marcote and Carbonell, 2000). Loss of function of *AtMPK4* in *Arabidopsis* caused the meiotic cytokinesis defect with drastically delayed or abortive mitosis and cytokinesis (Beck et al., 2011). MPK4 was phosphorylated and activated by MKK6/ANQ (Takahashi et al., 2010). Recently, Zeng et al. (2011) proved that *AtMPK4* played a specific role in meiotic cytokinesis during pollen development via the TES/STUD/AtNACK2–AtANPs–AtMKK6–AtMPK4 module.

A total of 20, 17 and 21 MAPK family genes have been identified from *Arabidopsis*, rice, and poplar genomes, respectively (Agrawal et al., 2003; Nicole et al., 2006; Rohila and Yang, 2007). As far as we know, only 3 SIMAPKs including *SIMAPK1*, *SIMAPK2*, and *SIMAPK3* were identified in tomato. To date, no systematic investigations of MAPK family proteins have been reported in tomato. The functional analysis of each MAPK family gene has not been performed, despite the importance of MAPK proteins in multiple aspects of tomato physiology. The recently completed Genome Sequencing Project for tomato (*Solanum lycopersicum* L.) provides an opportunity to deduce the tomato MAPK gene family and infer its evolutionary history. In the present study, we searched for all non-redundant sets of SIMAPK genes, and then obtained the full cDNA sequences of 13 newly identified SIMAPKs via database analysis and PCR-based approaches. The genomic structure, chromosomal location, and sequence homology of all SIMAPKs were then investigated, followed by comparative phylogenetic analysis, exon/intron mapping, and structural analysis of conserved protein motifs of the SIMAPK family genes. Finally, the temporal and spatial expression profiles of SIMAPK genes in different organs and tissues and in response to heat stress were investigated through RT-PCR and quantitative real-time PCR (QRT-PCR) analyses.

2. Materials and methods

2.1. Identification and isolation of tomato MAPKs

The *Arabidopsis* MAPK protein and cDNA sequences were downloaded from The Arabidopsis Information Resource (TAIR) (Garcia-Hernandez et al., 2002). To identify putative MAPKs in tomato (*S.*

lycopersicum L.), multiple database searches were performed, including National Centre for Biotechnology Information (NCBI; <http://www.ncbi.nlm.nih.gov/>), the SOL Genomics Network (SGN; <http://www.sgn.cornell.edu>) and tomato plant GDB database (<http://www.plantgdb.org>). After searching for MAPK gene family, DNASTAR and FGENESH (<http://linux1.softberry.com/berry>) were used to analyze and predict those MAPK domains. NCBI ORF finder was used to find putative open reading frames and functional domains were determined by blastp of NCBI.

The cDNA sequences of 13 newly identified SIMAPKs were amplified by PCR using primers designed based on their predicted sequences (listed in Supplemental Table 1). Total RNA was extracted from tomato (*S. lycopersicum* L. cv. Micro-Tom) young leaves using the TRIZOL reagent (Invitrogen, USA) according to the manufacturer's instructions. The first cDNA strand was generated following the manufacturer's protocol using Improm-TM Reverse Transcription system (Promega, Madison, USA). PCRs were performed on TGRADIENT Thermal Cycler machines (Biometra, Germany) as following: 94 °C for 3 min, followed by 35 cycles of 94 °C for 30 s, 50 to 58 °C (depending on the specific primers) for 30 s and 72 °C for 70 s, with a final 7 min elongation at 72 °C. The amplified cDNA fragments were cloned and sequenced. The whole target fragment was verified to be the full length cDNA by PCR with gene-specific primer sets and by searching SGN database and Kazusa Full-length Tomato cDNA Database (<http://www.pgb.kazusa.or.jp/kaftom/blast.html>). For *SIMAPK11*, *SIMAPK12*, *SIMAPK15* and *SIMAPK16*, the 5' and 3' rapid amplifications of cDNA end (RACE) were carried out using a SMART RACE cDNA Amplification Kit following the protocol recommended by the manufacturer (BD Bioscience Clontech, CA). RACE products were cloned and sequenced. Amplification, cloning, and sequencing were repeated to avoid PCR-induced errors in the sequence.

Finally, the ORFs of the 13 putative SIMAPKs were determined by ORF Finder (<http://www.ncbi.nlm.nih.gov/gorf/gorf.html>) and homologous alignment. ExPASy Proteomics Server (<http://www.expasy.org/tools/#primary/>) was used to identify their primary functional domains.

2.2. Gene positions on tomato chromosomes

The exon/intron structures for individual SIMAPK genes were determined by aligning the cDNA sequences to their corresponding genomic DNA sequences. To determine the locations of SIMAPKs genes on chromosomes, the SIMAPKs sequences were further used as query sequences for the BLASTN search against SGN Tomato Whole genome Scaffolds data (2.30) (<http://www.sgn.cornell.edu/tools/blast/>).

2.3. Cis-elements in promoter sequences analysis

The 3 kb of genomic sequences upstream of the 5'-UTR representing their promoter region was obtained for upstream cis-element analysis with PlantCare (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>).

2.4. Multiple-sequence alignments and phylogenetic analysis

Multiple-sequence alignments of the 16 SIMAPKs and 20 AtMAPKs were performed using the ClustalX program (version 1.83) (Thompson et al., 1997). Conserved motifs were identified from the multiple alignment analyses using Clustal W and MEME version 3.0. Phylogenetic trees were constructed using MEGA 4.1 program with the neighbor-joining (NJ) method (Saitou and Nei, 1987) and bootstrap analysis (1000 replicates).

2.5. Plant growth and heat-stress treatment

Tomato plants (cv. Micro-Tom) provided by Tomato Genetics Resource Center (University of California, Davis) were grown in growth

chambers under 18-h light ($300 \mu\text{mol photons m}^{-2} \text{s}^{-1}$) at 28°C and 6-h dark at 18°C until flowering.

The leaves, stems, roots, petals, stamens, pistils, and fruits were collected from flowering tomato plants. Flower buds were collected at 5 stages of floral development, which were roughly defined as the following: Bud I: 2–4 mm green bud (st8–11); Bud II: 5–7 mm green bud (st12–15); Flower I: flowers at the day of opening (st20); Flower II: flowers 2 d after opening, Flower III: flowers 3 d after opening (Brukhin et al., 2003). All the samples were immediately frozen in liquid nitrogen and stored at -75°C until RNA isolation.

For high temperature treatment, 28-day-old tomato seedlings were incubated at $45 \pm 1^\circ\text{C}$ in a growth chamber under continuous light of $300 \mu\text{mol m}^{-2} \text{s}^{-1}$. The young leaves were collected at 0 h, 0.25 h, 0.5 h, 1 h, 2 h, 4 h and 8 h after heat stress treatment, and stored at -75°C until RNA isolation.

2.6. RT-PCR and QRT-PCR analysis

Total RNA and the first strand of cDNA were prepared as described above. RT-PCR was carried out using the primer pairs for *SIMAPKs* and actin primer pairs (Supplemental Table 2). The conditions for RT-PCR were as following: 4 min at 94°C , followed by 30 cycles of 30 s at 94°C , 30 s at $55\text{--}58^\circ\text{C}$ (varying with specific primers), and 45 s at 72°C , and a final extension at 72°C for 7 min to complete the reaction.

QRT-PCR was carried out using the primer pairs listed in Supplemental Table 3. Specificity of each primer to its corresponding gene was checked using the BLASTN program of the NCBI. A sample of cDNA (1 μg) was subjected to real-time PCR in a final volume of 20 μl containing 12.5 μl SYBR Green Master Mix Reagent (Takara, Japan) and gene specific primers (3 pmol). QRT-PCRs were performed on a real-time PCR machine (STRATAGENE, MX3500), programmed to heat for 30 s at 95°C , followed by 40 cycles of 5 s at 95°C and 45 s at $55\text{--}58^\circ\text{C}$ (varying with specific primers), at the end, 1 cycle of 1 min at 95°C , 30 s at 50°C and 30 s at 95°C . Two biological and three technical replicates were performed for each sample. To normalize the total amount of cDNA present in each reaction, the tomato Ubi3 gene was co-amplified as an endogenous control for calibration of relative expression. The $2^{-\Delta\Delta\text{CT}}$ method of relative gene quantification recommended by Applied Biosystems (PE Applied Biosystems, USA) was used to calculate the expression level of different samples (Livak and Schmittgen, 2001).

3. Results

3.1. Isolation and identification of *SIMAPK* family genes in tomato

To identify *SIMAPK* family genes in tomato, BLAST searches against the SGN and GenBank databases were performed using *Arabidopsis* MAPK proteins as query sequences. A total of 18 unigenes and additional 3 genome genes were obtained from tomato unigene and genome databases, respectively, using the TBLASTN program. Different sequences present on the same or overlapping contigs in different databases were further identified and removed to obtain a set of non-redundant *SIMAPKs*. Each predicted *SIMAPK* sequence was confirmed by FGENSESH (<http://www.softberry.com/berry.phtml?topic=fgenesh>). These predicted amino acid sequences were analyzed by ExPASy Proteomics Server to find their conserved domains. The overall analysis revealed that the *SIMAPK* gene family comprises of 16 members in the tomato genome. In addition to the 3 previously reported *SIMAPK* genes, 13 novel *SIMAPK* genes were found (Table 1).

The full-length cDNA sequences of all 13 newly identified *SIMAPKs* were isolated through PCR-based approaches. The PCR primers (Supplemental Table 1) were designed based on the predicted sequences of 13 unigenes using FGENSESH. The cDNA and amino acid sequences of each *SIMAPK* have been submitted to the GenBank and were listed in Supplemental Fig. 1. The average length of *SIMAPK* full-length cDNAs was 1966 bp ranging from 1461 bp(*SIMAPK1*) to 2806 bp(*SIMAPK13*) (Supplemental Fig. 1). The open reading frame (ORF) length ranged from 1113 bp (*SIMAPK8*) to 1866 bp(*SIMAPK12*), with an average ORF length of 1329 bp. The ORFs encoded polypeptides of 370 aa to 621 aa, with a predicted molecular mass range of 32.52–67.33 kDa (Table 1).

3.2. Sequence alignments and phylogenetic analysis

Alignment of the predicted amino acid residues of *SIMAPKs* is shown in Fig. 1. All the *SIMAPK* proteins contain eleven domains (I–XI) (Fig. 1A) and a highly conserved region of about 300 amino acid residues in their N-terminal portion (Fig. 1B), which is similar to AtMAPKs (Fig. 1C). Domain VIII contains the threonine and tyrosine residues, named T-X-Y motif, that have to be phosphorylated for the activation of *SIMAPKs*. Interestingly, all the tomato *SIMAPK* protein sequences were found to contain four types of special

Table 1
Gene analysis of *SIMAPKs* in *Solanum lycopersicum*.

Name	Chromosomes ^a	Position ^b	Deduced polypeptide ^c			Types ^d	Group	Genebank accession number
			ORF length (bp) (aa)	(PI)	MW(KD)			
<i>SIMAPK1</i>	12	103,858,01–103,977,00	1191(396)	5.49	45.51	TEY	A	AY261512.1
<i>SIMAPK2</i>	8	435,410,1–437,750,0	1185(394)	5.53	45.23	TEY	A	AY261513.1
<i>SIMAPK3</i>	6	191,701–195,500	1122(373)	5.45	42.82	TEY	A	AY261514.1
<i>SIMAPK4</i>	11	52,944,501–52,949,400	1119(372)	5.01	42.72	TEY	B	HM367593
<i>SIMAPK5</i>	1	781,179,01–781,254,00	1122(373)	5.97	42.92	TEY	B	HM036493
<i>SIMAPK6</i>	5	58,963,701–58,969,000	1131(376)	6.76	43.08	MEY	B	JF791807
<i>SIMAPK7</i>	8	61,696,201–61,700,200	1140(379)	5.84	43.55	TEY	B	JF791808
<i>SIMAPK8</i>	2	42,513,001–42,517,600	1113(370)	8.28	42.68	TEY	C	HM189670
<i>SIMAPK9</i>	4	62,419,101–62,428,700	1119(372)	6.32	42.80	TEY	C	HM245320
<i>SIMAPK10</i>	10	1,832,401–1,837,100	1788(595)	9.25	67.33	TDY	D	JF795445
<i>SIMAPK11</i>	7	62,134,901–62,139,500	1479(492)	9.43	55.38	TDY	D	JF795446
<i>SIMAPK12</i>	7	61,554,301–61,559,600	1866(621)	9.18	70.55	TDY	D	JF795447
<i>SIMAPK13</i>	1	719,729,01–719,793,00	1377(458)	8.97	51.89	TDY	D	JF795448
<i>SIMAPK14</i>	4	138,930,1–139,580,0	1539(512)	6.59	56.56	TDY	D	JF795449
<i>SIMAPK15</i>	5	244,040,1–244,410,0	1239(412)	5.63	47.12	TDY	D	JF795451
<i>SIMAPK16</i>	12	402,811,01–402,890,00	1728(575)	6.49	65.63	TDY	D	JF795450

^a Chromosomal localization of the *SIMAPK* genes.

^b Position of *SIMAPKs* on the chromosomes.

^c Length of open reading frame in base pairs, the number of amino acids, molecular weight (kilodaltons), and isoelectric point (pI) of the deduced polypeptide.

^d Activation domain predicted by blastp of NCBI. TEY: Thr-Glu-Tyr, TDY: Thr-Asp-Tyr, MEY: Met-Glu-Tyr.

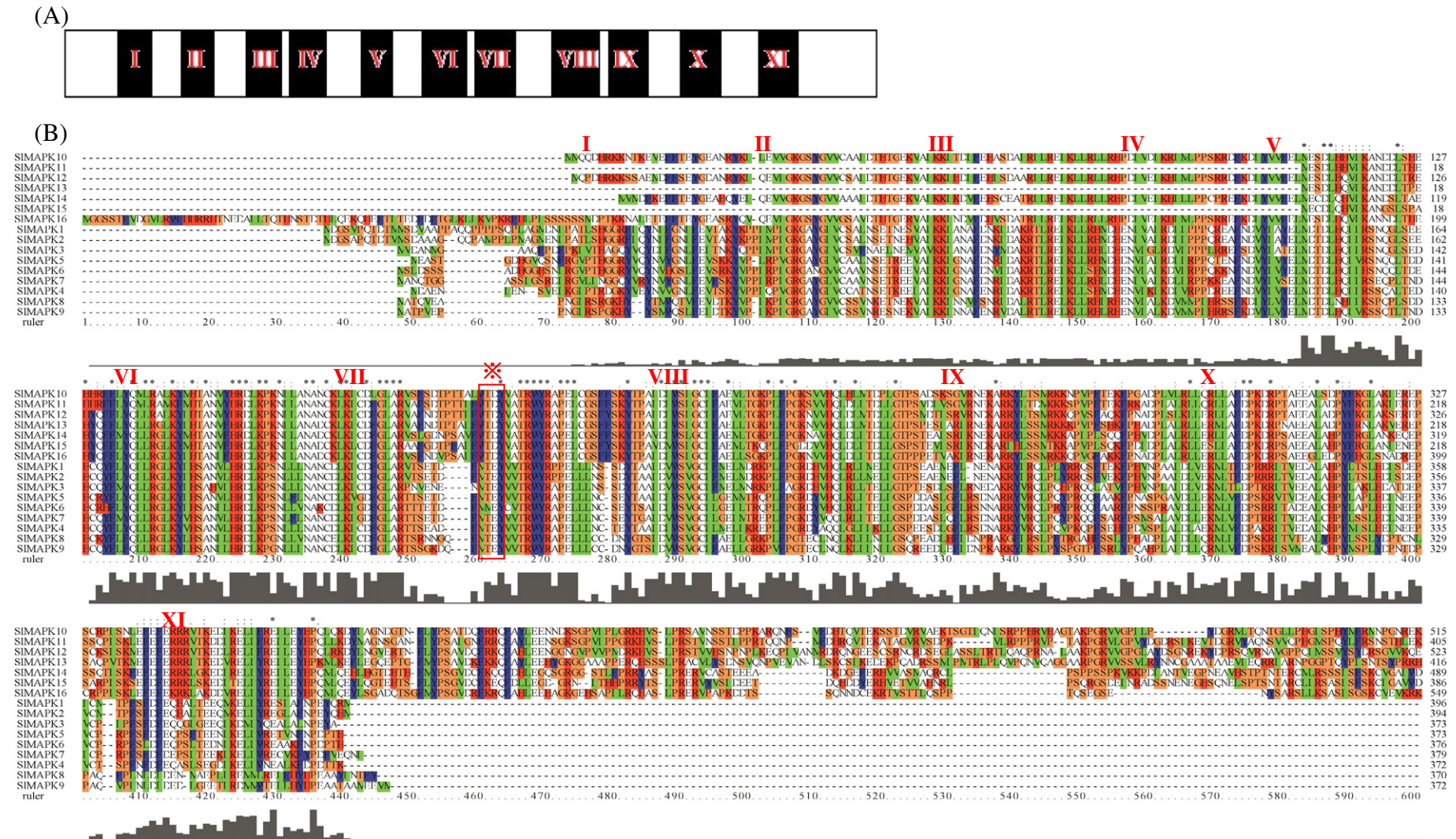


Fig. 1. (A) Eukaryotic MAP kinase primary structure: shaded areas represent regions containing the eleven domains (I–XI) that are found in all serine/threonine protein kinases; open areas indicate regions that are less conserved. (B) Comparison of the deduced amino acid sequences of SIMAPKs. Roman numerals indicate regions containing the eleven domains (I–XI) that are found in tomato serine/threonine protein kinases. The height of the gray columns represents the number of similar amino acid residues. Red box shows residues corresponding to the sites of regulatory phosphorylation. (C) Comparison of the deduced amino acid sequences of *Arabidopsis* AtMAPKs.

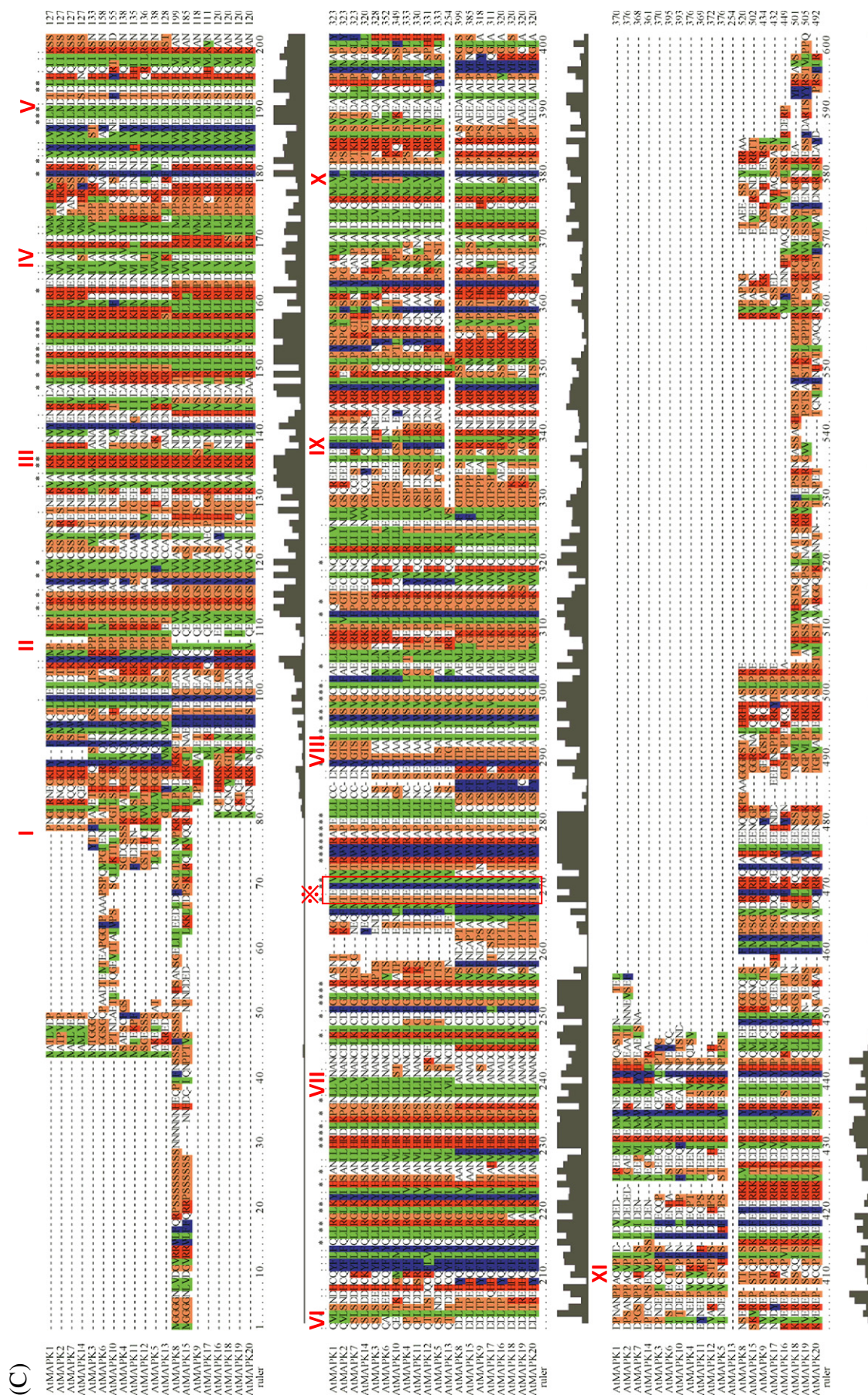


Fig. 1 (continued).

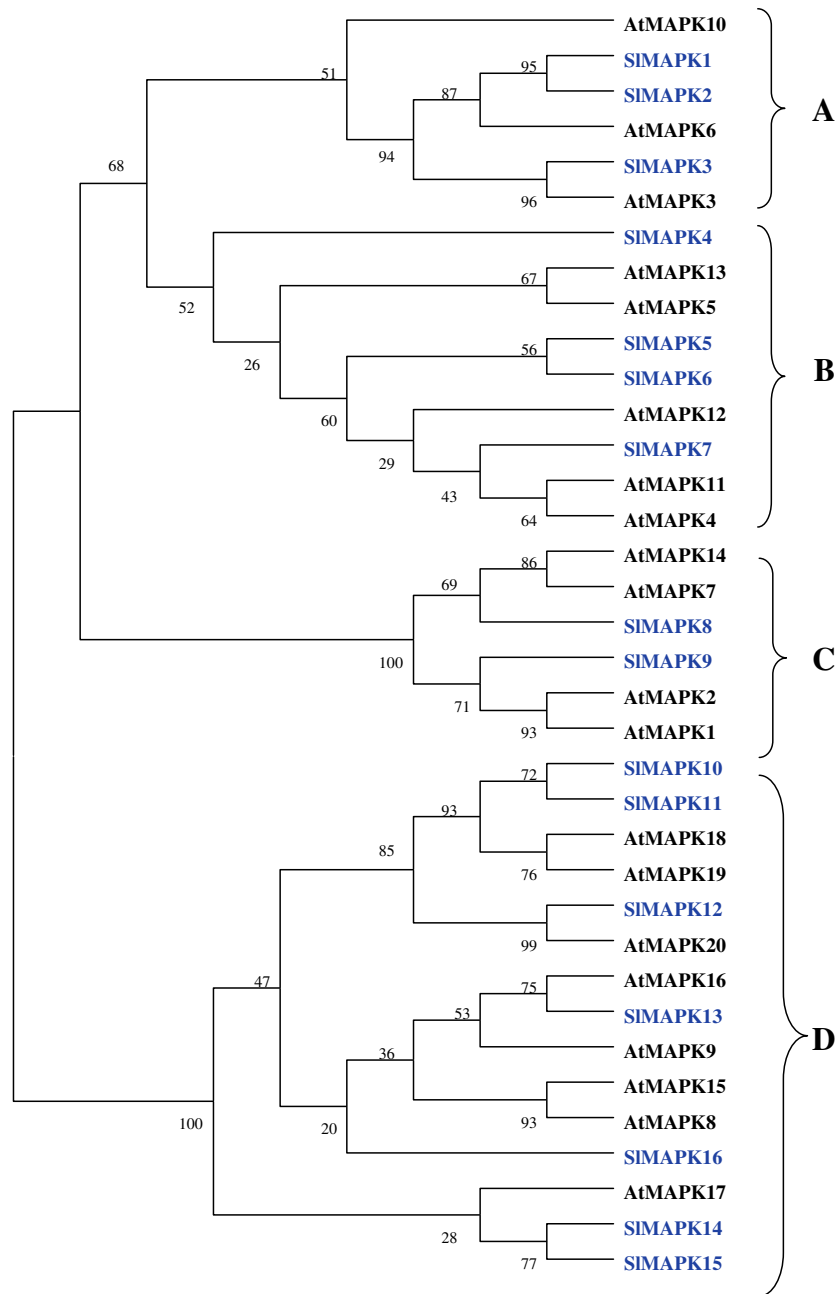


Fig. 2. Neighbor-joining phylogenetic tree of the MAPK family genes in tomato (*Solanum lycopersicum*, Sl) and *Arabidopsis thaliana* (At). The unrooted tree was generated using MEGA4.1 program with the neighbor-joining method. Bootstrap values (above 50%) from 1000 replicates are indicated at each branch.

subdomains: active site, ATP binding site, substrate binding site, and activation loop (A-loop) (Supplemental Fig. 2).

In general, we classified 16 SIMAPKs into four different groups based on their predicted amino acid sequences (Table 1, Supplemental Fig. 3) and phylogenetic analysis. SIMAPK1–4, SIMAPK5–8, SIMAPK9–10, and SIMAPK11–16 belong to groups A, B, C, and D, respectively. SIMAPKs belonging to groups A, B and C all possess a Thr-Glu-Tyr (TEY) activation domain except SIMAPK6, in which the Met-Glu-Tyr (MEY) sequence replaces the TEY domain, whereas those of group D possess a Thr-Asp-Tyr (TDY) activation domain (Fig. 1, Supplemental Fig. 1).

A phylogenetic tree was constructed for MAPKs from tomato and *Arabidopsis* (Fig. 2). The resulting tree showed almost the same topology as those formed by using the tomato or *Arabidopsis* alone (Fig. 2, Supplemental Fig. 3). Members belonging to the same groups are invariably clustered together.

3.3. Gene structure analysis and chromosomal distribution of SIMAPKs

A comparison of full-length cDNA sequences with the corresponding genomic DNA sequences revealed exon/intron structures for each individual SIMAPK gene. The number of introns varied from 1 (SIMAPK8 and SIMAPK9) to 10 (SIMAPK16) (Fig. 3). Within each group, exon/intron structures of different SIMAPKs were quite similar, further validating the classification of the SIMAPK family genes in this study.

Intron phase was also assessed for all SIMAPK gene models (Fig. 3). We found 33.6%, 35.8% and 30.4% of introns are within phases 0, 1 and 2, respectively, which is quite different from those observed for popular and *Arabidopsis* MAPK genes, of which the majority of introns are within phase 0 (Nicole et al., 2006).

The 16 SIMAPK genes were anchored on ten of the 12 tomato chromosomes as shown in Fig. 4. Two SIMAPKs were found on each of

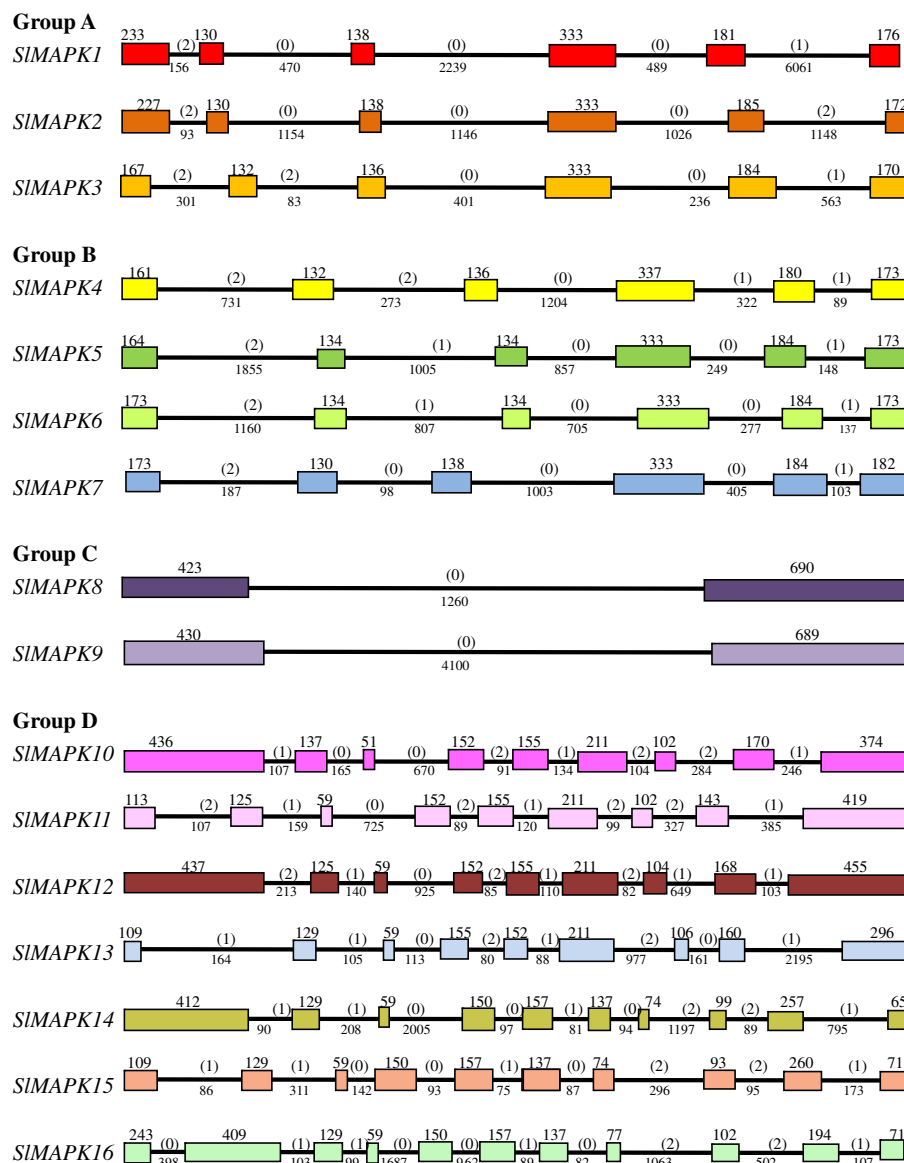


Fig. 3. Intron and exon organization of tomato *SIMAPK* genes. Introns and exons are represented by black lines and colored boxes, respectively. The length in base pairs of each intron and exon is also indicated. Numbers between brackets correspond to the intron phase. *SIMAPK*s have been grouped according to the phylogenetic classification.

chromosomes 1, 4, 5, 7, 8 and 12, while only one was found on each of the remaining chromosomes (Table 1; Fig. 4). Although paralogs such as *SIMAPK1* and *SIMAPK2*, *SIMAPK8* and *SIMAPK9*, *SIMAPK10* and *SIMAPK11* sharing high similarity of amino acid sequences based on the multiple sequence alignments (Fig. 1), they didn't form clusters and were distributed on different chromosomes (Fig. 4), indicating tandem duplication has not been involved in the evolution of *SIMAPK* genes.

3.4. Cis-elements in promoter sequences of *SIMAPK*s

To further understand transcriptional regulation and the potential function of *SIMAPK*s, the cis-elements in their promoter sequences were predicted. More than 270 cis-elements were found in the 3-kb upstream region of tomato *SIMAPK* genes using the PlantCare database (Table 2, Supplemental Fig. 4). All of the *SIMAPK*s contained Skn-1-motif required for endosperm expression. Most of them had the cis-regulatory elements involved in circadian regulation. HSE elements involved in heat stress response were found in the promoter regions of all the 16 *SIMAPK* genes except *SIMAPK6* and *SIMAPK16*.

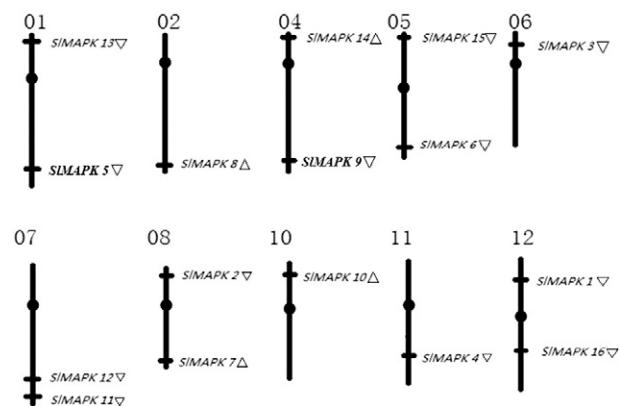


Fig. 4. Distribution of *SIMAPK* genes on tomato chromosomes. Black spots on the tomato chromosomes (vertical bar) indicate the position of centromeres. The triangles indicate the transcription direction. The chromosome numbers are indicated at the top of each bar.

Table 2The putative cis-elements with more than 6 bp identified in 16 *SIMAPK* genes in tomato.

Gene	Element ^a	Element ^b
<i>SIMAPK1</i>	CAT-box, CCGTCC-box, GCN4-motif, O ₂ -site, Skn-1-motif ⁴	ARE, EIRE, ERE, HSE ³ , MBS, SARE, TCA-element, TC-rich repeats
<i>SIMAPK2</i>	Circadian, GCN4-motif, O ₂ -site Skn-1-motif ³	CGTCA-motif, GARE-motif ⁴ , HSE, MBS, P-box, TCA-element ³ , TC-rich repeats ² , TCA-element ³
<i>SIMAPK3</i>	circadian ² , GCN4-motif, Skn-1-motif ²	ABRE ³ , ARE, CGTCA-motif, ERE, HSE, TCA-element ² , TC-rich repeats
<i>SIMAPK4</i>	as-2-box, circadian ³ , Skn-1-motif ²	CGTCA-motif, ERE, GARE-motif, HSE ² , MBS, TCA-element, TC-rich repeats, TGACG-motif
<i>SIMAPK5</i>	Circadian ² , GCN4-motif, Skn-1-motif ³	ARE, Box-W1, CGTCA-motif, HSE, MBS ² , TCA-element ² , TC-rich repeats ³ , TGACG-motif
<i>SIMAPK6</i>	CAT-box, circadian ² , GCN4-motif O ₂ -site, Skn-1-motif ³	ABRE ² , ARE, CGTCA-motif, ERE, LTR, P-box, TC-rich repeats, TCA-element ²
<i>SIMAPK7</i>	Circadian, GCN4-motif, Skn-1-motif ³	ARE ² , ERE, HSE ³ , LTR ² , TCA-element, TC-rich repeats ²
<i>SIMAPK8</i>	Circadian ² , O ₂ -site, Skn-1-motif	ABRE ⁴ , ERE, HSE ³ , TCA-element
<i>SIMAPK9</i>	as-2-box, O ₂ -site,	Box-W1, GARE-motif ² , MBS, TCA-element
<i>SIMAPK10</i>	as-2-box, circadian, HD-Zip1, HD-Zip2, O ₂ -site, Skn-1-motif ²	ABRE ³ , ARE ² , HSE ⁵ , TATC-box ² , TCA-element, TGACG-motif
<i>SIMAPK11</i>	Circadian ² , Skn-1-motif ²	ABRE, ARE ² , CGTCA-motif ⁴ , ERE, HSE, GARE-motif ² , LTR ² , TCA-element ² , TGACG-motif, MBS, TC-rich repeats ²
<i>SIMAPK12</i>	CCGTCC-box, Skn-1-motif ⁴ ,	ABRE, CGTCA-motif ² , ERE, HSE, GARE-motif ² , TC-rich repeats, TGACG-motif ²
<i>SIMAPK13</i>	Circadian ³ , Skn-1-motif ⁴	ABRE, ARE, CGTCA-motif, HSE, LTR, TC-rich repeats, TGA- element, TGACG-motif ²
<i>SIMAPK14</i>	as-2-box, circadian ³ , HD-Zip1, HD-Zip2, Skn-1-motif ⁴	ABRE ² , ARE, CGTCA-motif, MBS, WUN-motif
<i>SIMAPK15</i>	Circadian ² , Skn-1-motif	ABRE ² , ARE, AuxRR-core, CGTCA-motif ² , HSE ² , LTR, MBS, TCA-element ² , TC-rich repeats, TGACG-motif
<i>SIMAPK16</i>	Circadian ² , GCN4-motif,	ABRE ³ , ERE, LTR, TCA-element, TC-rich

^acis-elements involved in the growth and development in plant.^bcis-elements related to the stress response in plant.^cRed numbers represent the repeats (two or more than two) of each cis-element in a particular *SIMAPK* promoter, while the others only contain one copy of corresponding cis-element.

AACA-motif, endosperm-specific negative expression; ABRE, ABA-responsive element; ARE, anaerobic response element; As-2-box, involved in shoot-specific expression and light response; AuxRR-core, auxin-responsive element; Box-W1, fungal elicitor responsive element; CCGTCC-box, involved in meristem specific activation; CAT-box, element related to meristem expression; Circadian, circadian-control element; CGTCA and TGACG-motif, MeJA-responsive element; dOCT, related to meristem specific activation; EIRE and ELI-box3, elicitor-responsive element; ERE, ethylene-responsive element; GARE-motif and P-box, gibberellin-responsive element; GCN4-motif, involved in endosperm expression; HD-Zip1, involved in differentiation of the palisade mesophyll cells; HD-Zip2, involved in leaf morphology development; HSE, heat stress-responsive element; LTR, low temperature-responsive element; MBSI and MBSII, MYB binding site involved in flavonoid biosynthetic genes regulation; O₂-site, involved in zein metabolism regulation; RY-element, involved in seed-specific regulation; SARE and TCA-element, salicylic acid-responsive element; Skn-1-motif, a transcription factor that specifies mesodermal development; TATC-box, gibberellin-responsive element; TC-rich repeats, defense and stress responsive element; WUN-motif, wound-responsive element.

AuxRR-core sequences involved in auxin response were found in *SIMAPK7* and *SIMAPK8*. The LTR, a low-temperature responsive cis-element, was found in *SIMAPK6* and *SIMAPK16*. Interestingly, almost all of *SIMAPKs* contained one or more of cis-elements involved in signaling pathway of jasmonic acid-methyl ester (MeJA), ethylene (C2H4) and salicylic acid (SA). The cis-elements presenting in *SIMAPK*

promoter region revealed they might play an essential role in growth and development, and also in mediating responses to biotic and abiotic stresses.

3.5. Expression profiles of *SIMAPK* genes

Several genes of the *MAPK* family have been reported for their differential expression patterns in various plant tissues and organs. In the present study, the expression patterns of all *SIMAPK* genes in different tomato organs were characterized by RT-PCR (Fig. 5) and further investigated and confirmed by QRT-PCR (Fig. 6). We found that transcripts of most *SIMAPKs* could be detected in leaf, stem, flower and fruit using RT-PCR (Fig. 5). The quantitative analysis (QRT-PCR) also demonstrated that transcripts of all *SIMAPKs* could be detected in at least one tissue. Most *SIMAPKs* had higher expression levels in stamens or petals than in other organs, except that *SIMAPK9* exhibited a higher expression level in leaves.

During the tomato flower development, different members in the *SIMAPK* family exhibited distinct expression patterns (Fig. 7). The mRNA levels of *SIMAPK3* and *SIMAPK4* were significantly increased after the flower opening. The expression levels of *SIMAPK9* and *SIMAPK10* were significantly decreased at the early flower bud stage, while the expression of *SIMAPK8*, *SIMAPK14* and *SIMAPK16* was markedly decreased at the late flowering stage. *SIMAPK3*, *SIMAPK4* and *SIMAPK13* transcript levels were significantly increased until the flower II stage then markedly decreased. *SIMAPK2*, *SIMAPK11* and *SIMAPK13* exhibited a similar expression pattern, which was decreased gradually before the flower opening, and then suddenly increased after the flower opening (Flower II), but later decreased again. The expression levels of *SIMAPK1*, *SIMAPK5* and *SIMAPK12* seemed to have no obvious changes during tomato flower development.

The *SIMAPK* genes also exhibited differential expression patterns in response to high-temperature stress (45 ± 1 °C) (Fig. 8). Most *SIMAPKs* were up-regulated upon the exposure to the heat stress. The expression of *SIMAPK1*, *SIMAPK3*, and *SIMAPK8* gradually was increased in response to heat stress, while that of *SIMAPK2*, *SIMAPK4*, *SIMAPK5*, *SIMAPK6*, *SIMAPK7*, *SIMAPK9*, *SIMAPK12*, and *SIMAPK16* was increased at the early stage, then decreased at 0.5 h or 1 h after heat stress treatment. Although the mRNA transcript levels of *SIMAPK10*,

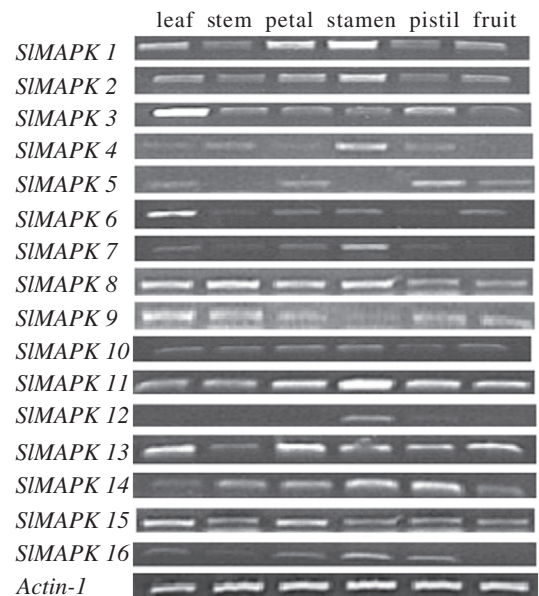


Fig. 5. Expression patterns of *SIMAPK* genes in different tomato organs characterized by RT-PCR.

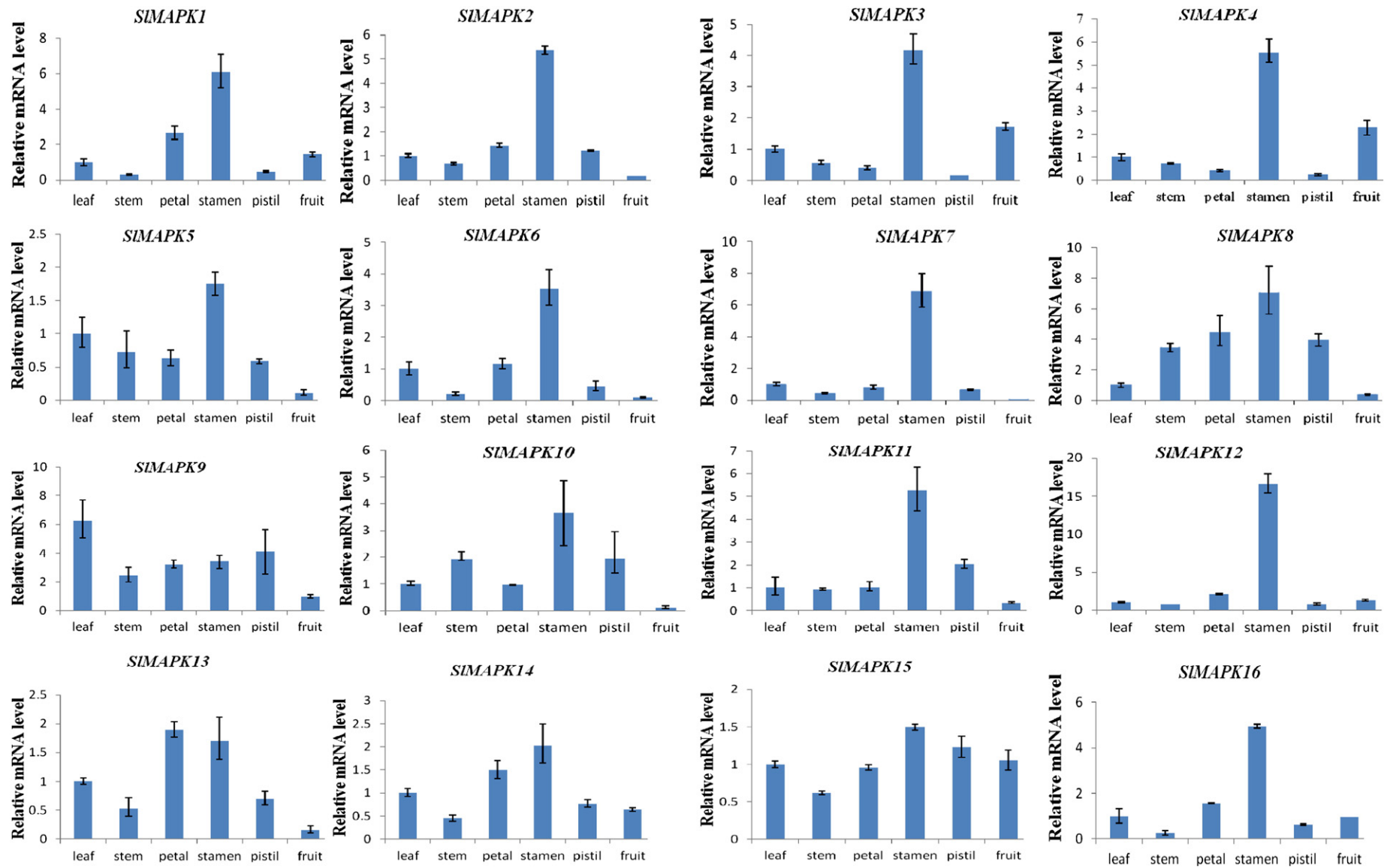


Fig. 6. Relative gene expression of SIMARK genes analyzed by QRT-PCR in different tomato organs (leaf, stem, petal, stamen, pistil, and fruit). Y-axes are scales of relative expression level (error bars indicate $\pm$ SD).

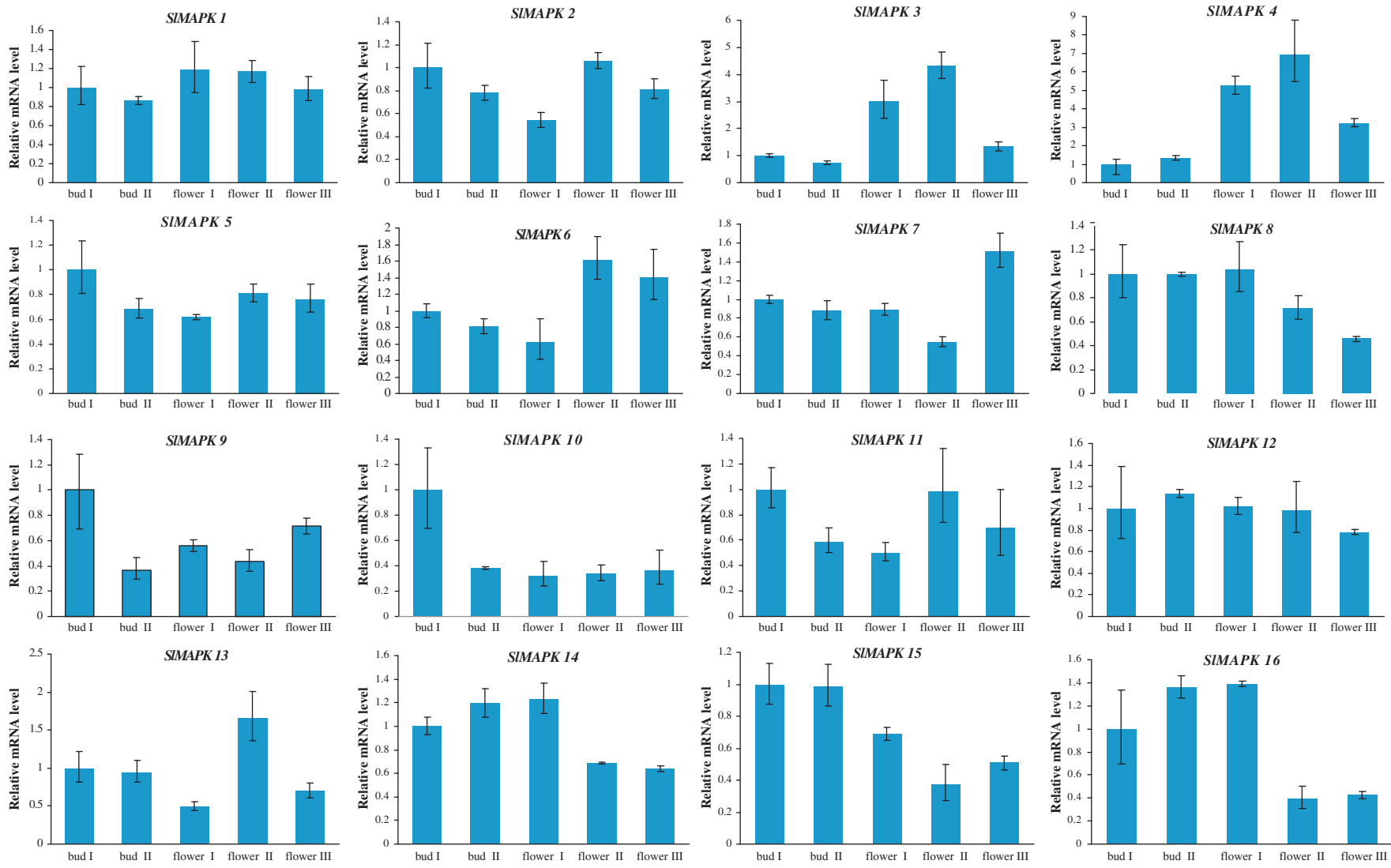


Fig. 7. Relative gene expression of *SIMAPK* genes analyzed by QRT-PCR in different stages of flower development. Bud I is the green bud about 2–4 mm; Bud II is the green bud about 5–7 mm; Flowers I, II and III are the whole flowers (excluding the sepals) on 0, 1 and 2 days after opening, respectively.

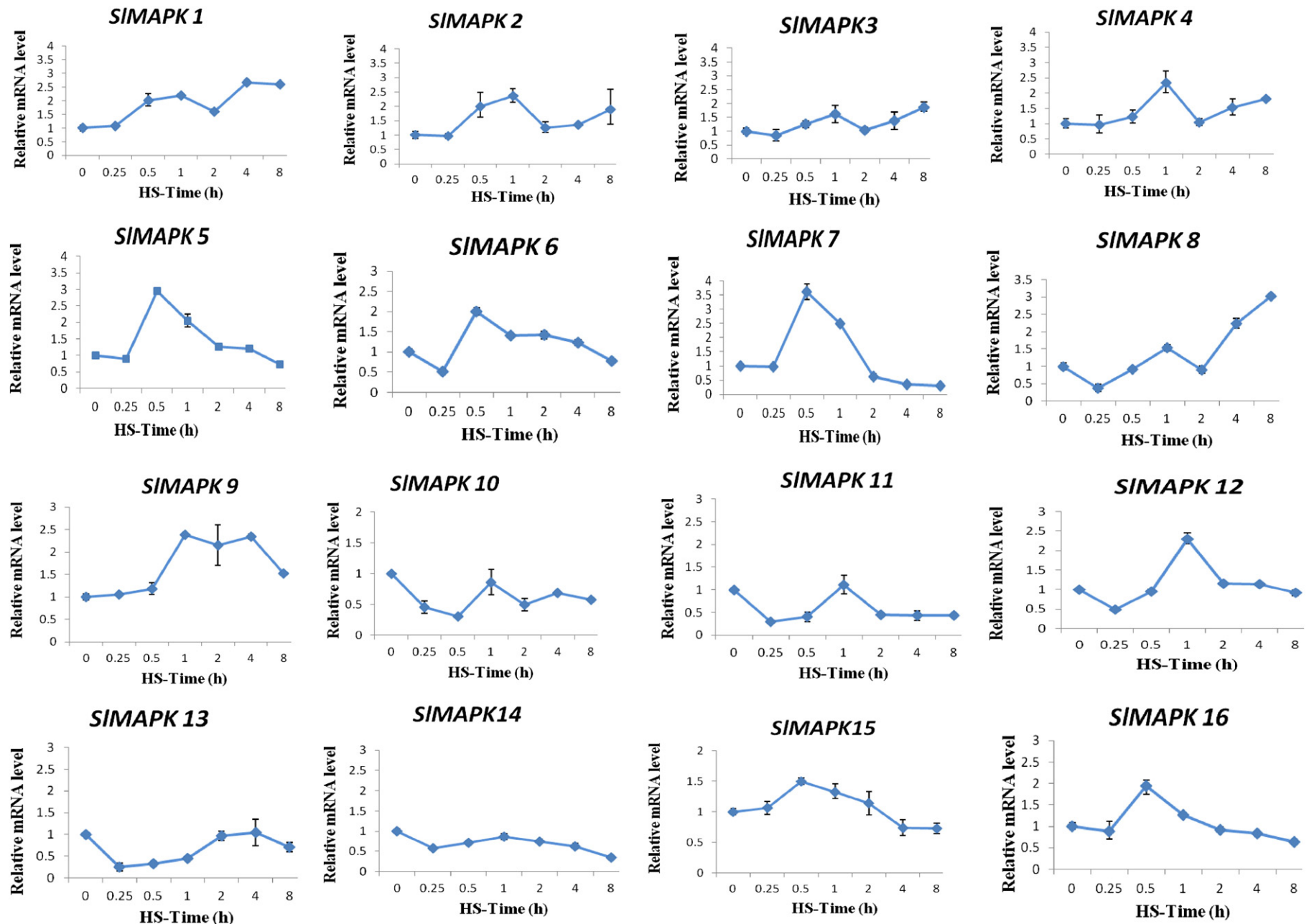


Fig. 8. Relative gene expression of SIMARK genes analyzed by QRT-PCR upon heat stress treatment. HS: high temperature treatment.

SIMAPK11, *SIMAPK13*, and *SIMAPK14* exhibited a little decrease after the heat shock, however, the changes were not significant.

4. Discussion

4.1. Characterization of *SIMAPKs* in tomato and evolution of *SIMAPK* genes

In plants, increasing evidence supports that MAPK pathways are involved in the regulation of development, growth and in responses to a diversity of biotic and abiotic stress. So far, 20, 17 and 21 *SIMAPK* genes in the *Arabidopsis*, rice and poplar genomes have been reported, respectively (Nicole et al., 2006; Reyna and Yang, 2006; Rohila and Yang, 2007). In this study, a total of 16 *SIMAPK* genes from tomato genome were identified and the full-length cDNA sequences of the 13 newly identified *SIMAPKs* were isolated. A T-X-Y domain containing the threonine and tyrosine residues typically found in all characterized MAPKs so far (Johnson and Lapadat, 2002; Krens et al., 2006) was also observed in the deduced *SIMAPKs* except *SIMAPK6*. *SIMAPK6* might represent a non-functional homolog because the Met-Glu-Tyr (MEY) sequence that replaces the TEY domain in this MAP kinase probably renders the protein inactive (Stulemeijer et al., 2007). All the deduced *SIMAPK* family genes in tomato were divided into four groups: A, B, C, and D according to the phylogenetic clades, which is similar to those in *Arabidopsis*, rice and poplar (Nicole et al., 2006; Reyna and Yang, 2006; Rohila and Yang, 2007). There are usually a number of conserved regions in the catalytic domain of protein kinases. Here, sequence similarity beyond the domains is quite high in most *SIMAPK* genes (Suppl. data Fig. S1). Indeed, in every plant species, MAPKs have either a TEY or TDY phosphorylation domain at the active site, and specifically TDY MAPKs also have long C-termini. TEY MAPKs have been extensively studied in several plants, such as *Arabidopsis*, tobacco, tomato, rice and Medicago (reviewed by Nakagami et al., 2005).

Phylogenetic analysis revealed that the organization of *Arabidopsis* and tomato *SIMAPK* proteins was very similar to each other in groups A, B, and C, implying that all *SIMAPKs* in these groups might derive from a common ancestor (Fig. 2). As previously observed in other MAPK groups in *Arabidopsis* and poplar, some members of the group D MAPK genes seemed to represent the paralogous products of recent genomic duplication events, since they possessed a very high degree of sequence similarity, and are located on different chromosomes (Hamel et al., 2006).

The results of structural analyses of tomato *SIMAPK* proteins will be helpful in their functional research. Comparative analysis of exon–intron junctions within the coding region of tomato MAPKs indicated that the numbers and sizes of the exons as well as the intron phases are similar for *SIMAPKs* within the same groups. Interestingly, members in each group share the same/similar numbers of intron with the corresponding group of *Arabidopsis* and poplar. Meanwhile, exon lengths in these three species are clearly more conserved than intron lengths. Similar gene structures between different species indicate the high conservation of these signaling components in all dicots beyond the primary sequence identity. The absence of duplicated gene clusters within the tomato MAPK gene families is similar with other plant species including *Arabidopsis*, rice and poplar (Hamel et al., 2006).

4.2. Diverse cis-elements in the promoters of *SIMAPK* genes

Studies on cis-elements can provide a crucial foundation for further functional dissection of the tomato *SIMAPK* genes. In the present study, we found that the upstream regions of all *SIMAPK* genes contain two or three cis-elements related to stress responses (Table 2), including ABRE (ABA response element), ERE (ethylene-responsive element), HSE (heat stress response element), LTR (low-temperature response element), SARE and TCA-element (SA response element), P-

box and TATC-box (MeJA-response element). These cis elements may play important roles in regulating gene expression in stress response. Our QRT-PCR results also proved that most of the *SIMAPKs* with HSE elements were up-regulated by heat stress treatment. Thus the existence of these cis-elements in tomato *SIMAPK* genes and their likely roles in regulating gene expression suggest that *SIMAPKs* may participate in various stress responses in tomato (Higgins et al., 2006).

4.3. Expression profiles of *SIMAPK* genes

In general, the expression levels of the majority of *SIMAPK* genes are relatively lower in the vegetative organs such as leaf and stem, and higher in the reproductive organs such as stamen and petal. More importantly, most *SIMAPK* genes exhibit different expression patterns during tomato flower development. Thus these genes are likely relevant to the flower or pollen development. *SIMAPK6* and *SIMAPK7* transcription levels are increased at the last period during the flower development, indicating they might participate in the flower aging. Many MAPK genes in *Arabidopsis*, tobacco and poplar have been proven to be differentially expressed in diverse tissues. *AtMPK9* was proven to be preferentially expressed in guard cells and positively regulate ROS-mediated ABA signaling in *Arabidopsis* (Jammes et al., 2009), *PtMAPK16-1* in poplar is highly expressed in both male and female floral bud, while *PtMAPK16-2* are barely detectable in these organs (Nicole et al., 2006). In tobacco, *NtSIPK* is expressed in leaf, stem and pollen, while the expression of *Ntf4* was found to be restricted to seed, pollen and anther (Voronin et al., 2001, 2004). Differential spatial-temporal transcript accumulation patterns for most members of *SIMAPK* gene family indicated the functional divergence among *SIMAPK* genes during plant development and growth.

In recent years, increasing evidence has indicated that the MAPK cascade is involved in biotic and abiotic stress response. The best-characterized MAPKs are MPK3, MPK4 and MPK6 in *Arabidopsis*, all of which are activated by a diversity of stimuli including abiotic stresses, pathogens and oxidative stress (Pitzschke et al., 2009). MPK3, MPK4 and MPK6 in *Arabidopsis* are activated in the signaling process involved in salinity, osmotic, oxidative stress tolerance, which are also stimulated by low temperatures, and even by touch and wounding (Mizoguchi et al., 1996; Ichimura et al., 2000; Droillard et al., 2002). However, to date there are very few reports on the functional roles of other MAPKs in biotic and abiotic stresses, which is partially attributed to a lack of appropriate tools such as specific antibodies to detect low protein amounts or kinase activities.

It has been suggested that MAPK proteins classified in the same groups may have the similar functions (Nicole et al., 2006). AtMAPKs of group A have been described to be positive regulators of defense signaling (Romeis et al., 2001). The *SIMAPK1*, 2 and 3 belonging to group A in tomato are activated upon stress responses caused by the wound signaling peptide systemin, oligosaccharide elicitors, ultraviolet-B radiation and the fungal toxin fusaric acid (Higgins et al., 2006). Furthermore, *SIMAPK2* and *SIMAPK3* are activated in a Pto-specific manner upon expression of AvrPto and AvrPtoB (Pedley and Martin, 2004). *AtMAPK3* and *AtMAPK6* have been reported to regulate innate immunity in *Arabidopsis thaliana* (Nakagami et al., 2005). *SIMAPK1*, 2, and 3 play an important role in the resistance response of tomato (Stulemeijer et al., 2007). The phylogenetic tree (Fig. 2) indicates that *AtMAPK3*, 6 and *SIMAPK 1, 2, 3* have close phylogenetic relationship. So we can infer the potential functions of *SIMAPKs* from the known AtMAPKs in *Arabidopsis*. MPK3 and MPK6 in *Arabidopsis* were involved in the early anther development (Hord et al., 2008). In the present study, we found that *SIMAPK3* was expressed at a markedly higher level in stamen than other organs, indicating that *SIMAPK3* might also regulate early anther development. Efforts to identify single and multiple knock-out lines for these kinases would help to prove the evolution of protein functions (Colcombet and Hirt, 2008).

5. Conclusion

Taken together, 16 putative *SIMAPK* genes in tomato were investigated, which are comparable to those in *Arabidopsis*. The full cDNA sequences of 13 novel *SIMAPKs* were determined. The *SIMAPK* family genes were classified into four groups, resulting in a high level of functional divergence. *SIMAPK* genes showed different temporal and spatial patterns of expression during flower development or under heat stress. Interestingly, most *SIMAPK* genes were expressed at higher levels in stamens than in other organs. Most *SIMAPK* genes were significantly up-regulated by heat stress treatment. The preferential expression of *SIMAPK* genes provides a new avenue for functional analyses in tomato. The major challenge for the future is to understand the specific role of the individual *SIMAPK* gene during plant growth and development and under different abiotic and biotic stresses.

Accession numbers

Sequence data from this article can be found in the GenBank/EMBL data libraries under the following accession numbers: [HM367593](#)(*SIMAPK4*), [HM036493](#)(*SIMAPK5*), [JF791807](#) (*SIMAPK6*), [JF791808](#)(*SIMAPK7*), [HM189670](#)(*SIMAPK8*), [HM245320](#)(*SIMAPK9*), [JF795445](#) (*SIMAPK10*), [JF795446](#)(*SIMAPK11*), [JF795447](#)(*SIMAPK12*), [JF795448](#)(*SIMAPK13*), [JF795449](#) (*SIMAPK14*), [JF795451](#)(*SIMAPK15*), [JF795450](#)(*SIMAPK16*).

Supplementary materials related to this article can be found online at [doi:10.1016/j.gene.2012.01.048](https://doi.org/10.1016/j.gene.2012.01.048).

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